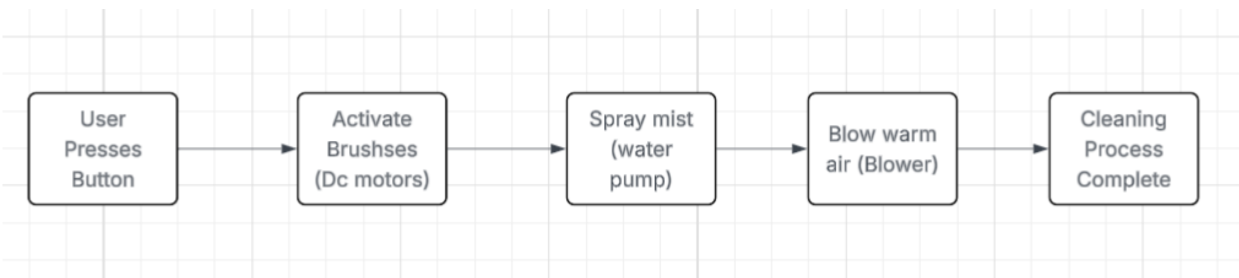
Automated Shoe & Foot Cleaning Pedestal Control System

In many public areas such as markets, schools, and transport terminals, people's shoes, slippers, or even bare feet easily get dirty due to mud, dust, and wet surfaces. Cleaning footwear manually with brushes or cloth requires time, effort, and bending down, which is inconvenient for the elderly or those who are in a hurry. A safe, automated shoe and foot cleaning system can make this task easier by removing dirt quickly without manual scrubbing. This would improve cleanliness, hygiene, and comfort for all users.

Dirty footwear and feet are a daily issue, especially in crowded and wet environments. Traditional shoe cleaners are designed only for shoes and may be harmful if used barefoot or with slippers. There is a need for a safe and automated system that can clean footwear or feet without causing injury and without requiring much physical effort from the user.



Open loop diagram

The system will be an **open-loop control system**. Once the user presses the button, the system runs a timed sequence of brushing, spraying, and drying. This design is simpler and safer because the user remains in control, and no complex sensors are needed. However, it could be upgraded to a closed-loop system in the future by adding dirt sensors or motion detectors to adjust cleaning automatically

Components:

- **Input:** A hand-pressed switch, will initiate the cleaning process.
- **Controller:** A microcontroller such as an Arduino or ESP32 will manage the cleaning cycle, activating the brushes, sprayer, and dryer in sequence.
- **Output:** The system produces clean shoes, slippers, or feet by brushing away dirt, spraying water/soap mist, and blowing warm air for drying.

To implement this automated shoe and foot cleaning pedestal, I would use an Arduino Uno or ESP32 as the controller. A waterproof start button would allow the user to activate the system easily. Once pressed, the microcontroller would power soft rotating silicone brushes driven by DC motors, a mist sprayer using a small water pump, and a gentle warm-air blower for drying. The brushes would be designed to be safe for both shoes and bare skin, while the mist spray ensures proper cleaning without discomfort.

The pedestal would be constructed with waterproof housing and proper drainage to avoid water buildup. All electronics would be splash-proof and operate at a safe low voltage (12V). The main challenges would be ensuring safety for barefoot users, preventing excess water splashing, and balancing cleaning strength with comfort. Testing would involve different shoe materials, slippers, and bare feet to verify that the system cleans effectively while remaining gentle. Once validated, this system could be installed in public markets, schools, and bus terminals, improving hygiene and reducing the effort needed for manual cleaning.